AIML INTERNSHIP — WEEK 1

Prompt Engineering Challenge

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This document presents five prompt engineering scenarios. For each scenario, the prompt written, the full AI response, and an explanation of why the prompt worked effectively are included.

Scenario 1: Summarising Notes

Scenario: You have 10 pages of handwritten class notes on Machine Learning and need a concise summary.

Prompt Written:

You are an expert academic tutor. I have the following notes on Machine Learning:
[Notes: Supervised learning uses labelled data. Unsupervised learning finds patterns in unlabelled data. Reinforcement learning uses rewards. Common algorithms: Linear Regression, Decision Trees, K-Means, Neural Networks.]

Please summarise these notes in 5 clear bullet points suitable for exam revision. Use simple language and highlight the most important concepts.

AI Response:

Here is a concise summary of your Machine Learning notes:

1. Supervised Learning: Trains on labelled data to predict outcomes. Examples: Linear Regression (predicts numbers), Decision Trees (classifies categories).

2. Unsupervised Learning: Finds hidden patterns in unlabelled data. Example: K-Means Clustering groups similar data points.

3. Reinforcement Learning: An agent learns by receiving rewards for correct actions and penalties for wrong ones — like training a dog with treats.

4. Key Algorithms: Linear Regression, Decision Trees, K-Means Clustering, and Neural Networks are the foundational ML algorithms to know.

5. Core Idea: ML systems improve their performance automatically as they process more data — no manual rule updates needed.

Why This Prompt Worked:

This prompt worked well because it assigned a specific role ('expert academic tutor'), provided the actual content to summarise, specified the exact output format (5 bullet points), defined the target audience (exam revision), and set a language constraint (simple language). These elements guided the AI to produce focused, useful output.

Scenario 2: Generating a Study Plan

Scenario: You have a major AI exam in 2 weeks and need a structured daily study plan.

Prompt Written:

Create a detailed 2-week study plan for an AIML internship student preparing for a final exam covering: AI Basics, Machine Learning, Deep Learning, Python Programming, and Prompt Engineering.

Requirements:
- Study sessions of 2 hours per day
- Include revision days and mock tests
- Format as a day-by-day schedule (Day 1 to Day 14)
- Add one motivational tip at the end

AI Response:

2-Week AIML Exam Study Plan:

Week 1 - Foundation Building:
Day 1: AI Basics - definitions, types, history (2 hrs)
Day 2: AI vs ML vs DL vs LLMs - differences and relationships (2 hrs)
Day 3: Python Basics - variables, loops, functions (2 hrs)
Day 4: Python Practice - dictionaries, file I/O, JSON (2 hrs)
Day 5: Machine Learning - supervised, unsupervised, reinforcement (2 hrs)
Day 6: ML Algorithms - Linear Regression, Decision Trees, K-Means (2 hrs)
Day 7: Week 1 Revision + Mini Mock Test (2 hrs)

Week 2 - Advanced Topics and Revision:
Day 8: Deep Learning - neural networks, layers, activation functions (2 hrs)
Day 9: LLMs and Transformers - GPT, BERT, Claude overview (2 hrs)
Day 10: Prompt Engineering - zero-shot, few-shot, role-based prompts (2 hrs)
Day 11: APIs and JSON - fetching data, parsing responses (2 hrs)
Day 12: Full Revision - AI + ML + Deep Learning (2 hrs)
Day 13: Full Revision - Python + APIs + Prompt Engineering (2 hrs)
Day 14: Final Mock Exam + Review Weak Areas (2 hrs)

Motivational Tip: Small daily progress beats last-minute cramming. Trust the process and show up every day!

Why This Prompt Worked:

This prompt was effective because it specified the exact topics to cover, set clear constraints (2 hours/day, 14 days), requested a specific format (day-by-day schedule), and asked for an extra motivational element. Detailed constraints prevent vague responses and produce actionable, structured output.

Scenario 3: Creating Interview Questions

Scenario: You are preparing for a junior AI/ML developer interview and need practice questions.

Prompt Written:

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You are a senior AI/ML interviewer at a top tech company. Generate 10 interview questions
for a Junior AI/ML Developer role.

Include:
- 4 conceptual questions (AI/ML theory)
- 3 Python coding questions
- 2 practical scenario-based questions
- 1 prompt engineering question

For each question, provide a brief expected answer (2-3 sentences).
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AI Response:

Conceptual Questions:

1. What is overfitting and how do you prevent it?
Expected: Overfitting occurs when a model learns training data too well and performs poorly on new data. Prevent it using regularisation, dropout, cross-validation, or collecting more training data.

2. Explain the difference between classification and regression.
Expected: Classification predicts discrete categories (spam/not spam). Regression predicts continuous values (e.g. house price).

3. What is a neural network and how does it learn?
Expected: A neural network is a system of interconnected layers of nodes. It learns by adjusting weights using backpropagation and gradient descent.

4. What is the role of a loss function in ML?
Expected: A loss function measures the gap between predicted and actual values. The model minimises this value during training.

Python Questions:

5. Write a Python function to find the second largest number in a list.
6. How do you read and parse a JSON file in Python?
7. Write a Python script to fetch data from an API using the requests library.

Scenario Questions:

8. Your model has 95% training accuracy but 60% test accuracy. What would you do?
Expected: This indicates overfitting. I would collect more data, apply regularisation, simplify the model, or use cross-validation techniques.

9. How would you approach building a chatbot for a college helpdesk?
Expected: Define intents, collect FAQ data, train an NLP model or use an LLM API, build a Python backend, and deploy with a simple web interface.

Prompt Engineering Question:

10. Write a prompt to make an LLM explain quantum computing to a 10-year-old.
Expected: 'Explain quantum computing in very simple words as if talking to a 10-year-old child. Use a fun analogy involving toys or games.'

Why This Prompt Worked:

This prompt used role assignment ('senior interviewer'), specified the exact breakdown of question types with counts, and requested expected answers. These elements together produced a well-structured, comprehensive, and immediately usable set of interview questions.

Scenario 4: Writing a Resume Summary

Scenario: You need a strong professional summary for your resume as an AIML intern.

Prompt Written:

Write a professional resume summary for the following candidate:

Name: Shaik Arshiya Tabasum
Role Applying For: Junior AI/ML Developer Intern
Skills: Python, Machine Learning basics, Prompt Engineering, API integration, JSON
Education: Currently pursuing B.Tech / BCA (AIML stream)
Strengths: Fast learner, team player, passionate about AI

Requirements:
- 3-4 sentences only
- Professional and confident tone
- Highlight enthusiasm and key skills
- Suitable for a fresher/intern level

AI Response:

Resume Summary:

Motivated and enthusiastic AI/ML intern with foundational expertise in Python programming, machine learning concepts, and prompt engineering. Experienced in building Python-based projects including smart assistants and API integration tools. Passionate about leveraging artificial intelligence to solve real-world problems, with a strong ability to learn new technologies quickly. Seeking an opportunity to apply and expand my skills in a dynamic, innovation-driven AI environment.

Why This Prompt Worked:

This prompt worked because it provided complete, structured candidate information, set precise length constraints (3-4 sentences), defined the tone (professional and confident), and specified the experience level (fresher/intern). Giving the AI all relevant context prevents assumptions and results in a personalised, ready-to-use resume summary.

Scenario 5: Creating a Lesson Plan

Scenario: You need to create a lesson plan for teaching AI basics to beginners in one hour.

Prompt Written:

You are an experienced AI educator. Create a 1-hour lesson plan to introduce Artificial Intelligence to complete beginners (college students with no prior AI knowledge).

Include:
- Clear time breakdown (e.g., 0-10 mins: Introduction)
- Learning objectives at the start
- At least one interactive activity
- A Q and A session
- A summary and takeaway at the end
- Teaching style: engaging, simple, and example-driven

AI Response:

1-Hour AI Basics Lesson Plan for Beginners:

Learning Objectives: By the end of this session, students will be able to define AI, distinguish it from ML and Deep Learning, and identify 3 real-world AI applications.

Time Breakdown:
0-5 min: Icebreaker - Name an AI tool you used this week (builds engagement)
5-15 min: What is AI? Definition, history, types with real-life examples
15-25 min: AI vs ML vs Deep Learning - simple visual diagram explanation
25-35 min: Real World AI Applications - healthcare, music, maps, social media
35-45 min: Interactive Activity - Students guess 'AI or Not AI?' for 10 scenarios
45-50 min: Introduction to Prompt Engineering - how to talk to AI effectively
50-55 min: Open Q and A Session - address student questions
55-60 min: Summary and Takeaway - 3 things to remember + free resources shared

Key Takeaway: AI is not magic — it is mathematics + data + clever programming. And you can learn it starting today!

Why This Prompt Worked:

This prompt assigned a specific educator role, set a clear time constraint (1 hour), defined the audience (complete beginners), and listed all required elements as a checklist. Using a checklist in the prompt ensures all requested components appear in the output. Specifying the teaching style shaped the tone of the entire lesson plan.